

Equation of a plane

Basic skills

$$Q1. \quad \underline{n} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \times 2 - 2 \times 1 \\ 2 \times 1 - 3 \times 2 \\ 3 \times 1 - -1 \times 1 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \\ 4 \end{pmatrix}$$

Could use $\rightarrow$

$$\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & -1 & 2 \\ 1 & 1 & 2 \end{vmatrix} = \underline{i} \begin{vmatrix} -1 & 2 \\ 1 & 2 \end{vmatrix} - \underline{j} \begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix} + \underline{k} \begin{vmatrix} 3 & -1 \\ 1 & 1 \end{vmatrix} \\ = -4\underline{i} - 4\underline{j} + 4\underline{k}$$

Also any scale is fine e.g. $\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$

$$Q2. \quad (\underline{a}, 3, 5), (\underline{b}, 3, 8), (\underline{c}, 1, 10)$$

$$a) \quad \underline{d}_1 = \underline{b} - \underline{a} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$$

$$\underline{d}_2 = \underline{c} - \underline{a} = \begin{pmatrix} -1 \\ -2 \\ 5 \end{pmatrix}$$

$$b) \quad \underline{n} = \underline{d}_1 \times \underline{d}_2 = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \times \begin{pmatrix} -1 \\ -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \times 5 - 3 \times -2 \\ 3 \times -1 - 1 \times 5 \\ 1 \times -2 - 0 \times -1 \end{pmatrix} = \begin{pmatrix} 6 \\ -8 \\ -2 \end{pmatrix}$$

c) $(5 - (\frac{1}{5})) \cdot (-\frac{6}{-2}) = 0$

d) $\underline{n} = (6, -8, -2)$ so

$$6x - 8y - 2z = d$$

Applying $(x, y, z) = (1, 3, 5)$

$$6 - 24 - 10 = d \quad d = -28$$

$$6x - 8y - 2z = -28$$

e) sub in $x=3, y=3, z=11$

$$6 \times 3 - 8 \times 3 - 2 \times 11 = 18 - 24 - 22 \\ = -28$$

so yes as $d = -28$

f) $x=0, y=0, z=0$

$$0 \neq -28$$

so origin is not in the plane

Competency skills

Q3. As $\underline{n} = (1, 0, -3)$

equation $1x + 0y - 3z = d$

sub $(0, -3, -2) \Rightarrow 0 - 3(-2) = d \Rightarrow d = 6$

so $x - 3z = 6$

Q4. $2x + 5y + 3z = 60$

a) $\underline{n} = (2, 5, 3)$ so $|\underline{n}| = \sqrt{2^2 + 5^2 + 3^2} = \sqrt{38}$

so $\hat{\underline{n}} = \frac{1}{\sqrt{38}} \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix}$ or $-\frac{1}{\sqrt{38}} \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix}$

b) Easy point on plane $y = z = 0$ so
 $2x = 60 \quad x = 30$

so $\left(\underline{r} - \begin{pmatrix} 30 \\ 0 \\ 0 \end{pmatrix} \right) \cdot \frac{1}{\sqrt{38}} \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix} = 0$

Q5. $\underline{a} = (4, 5, 1)$, $\underline{b} = (2, 3, 0)$, $\underline{c} = (-1, -1, -1)$

$$\underline{d}_1 = \underline{b} - \underline{a} = \begin{pmatrix} -2 \\ -2 \\ -1 \end{pmatrix} \quad \underline{d}_2 = \underline{c} - \underline{a} = \begin{pmatrix} -3 \\ -4 \\ -1 \end{pmatrix}$$

so $\underline{r} = \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ -4 \\ -1 \end{pmatrix}$

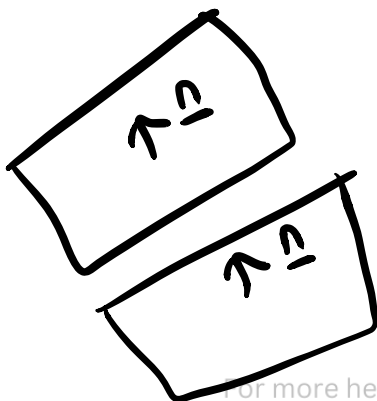
If $\underline{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ need
$$\begin{aligned} 4 - 2\lambda - 3\mu &= 0 \\ 5 - 2\lambda - 4\mu &= 0 \\ 1 - \lambda - \mu &= 0 \end{aligned}$$

First two eq's $\Rightarrow \mu = 1 \Rightarrow \lambda = \frac{1}{2}$

Then third eq is it consistent $-\frac{1}{2} \neq 0$
so NO SOL, NO intersection

Alternatively calculate $\underline{n} = \underline{d}_1 \times \underline{d}_2$
write cartesian and sub $(0, 0, 0)$

Q6.



Normals are Parallel.

Q7. $\pi_1: 2x - 4y + z = 15$

so π_2 being parallel $\underline{n} = (2, -4, 1)$

so $\pi_2: 2x - 4y + z = d$

going through $(5, 2, 10)$ so

$$10 - 8 + 10 = d \quad d = 12$$

$$\pi_2: 2x - 4y + z = 12$$

Advanced skills

Q8. $(\underline{r} - \underline{a}) \cdot \underline{n} = 0 \Rightarrow \underline{r} \cdot \underline{n} = \underline{a} \cdot \underline{n}$

$\underline{r} \cdot \underline{n} = ax + by + cz$ so $\underline{a} \cdot \underline{n} = d$

Q9. $(1, 0, 1)$

Point

$$\frac{x-1}{2} = \frac{y+3}{2} = z-1$$

line

Get a point from line $(1, -3, 1)$

and direction $\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$

Then need one more vector

$$\begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} \text{ use } \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \text{ for ease}$$

$$\text{So } \underline{d}_1 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \quad \underline{d}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\text{Then normal } \underline{n} = \underline{d}_1 \times \underline{d}_3 = \begin{pmatrix} 0 & -1 \\ 0 & -0 \\ 2 & -0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$

$$\text{So } -x + 2z = d$$

$$\text{Point } (1, 0, 1) \text{ so } -1 + 2 = d \Rightarrow d = 1$$

$$-x + 2z = 1$$

Q10. As they are colinear (i.e. form a line) / directions are parallel.
so they form a line

$$\text{Q11. } \underline{d}_1 = \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} \quad \underline{d}_2 = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$$

$$\text{So } \underline{r} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$$

other [→] other forms by finding normal

Q12. $2x + 5y + z = 10$

$$y = x + 4 = 2z$$

$$2x + 5(x+4) + \frac{1}{2}(x+4) = 10$$

$$7\frac{1}{2}x + 22 = 10 \Rightarrow x = -\frac{12}{\frac{15}{2}} = -\frac{24}{15}$$

so $x = -\frac{8}{5} \Rightarrow y = \frac{12}{5} \quad z = \frac{6}{5}$

$$\frac{1}{5}(-8, 12, 6)$$